

This is CS50

Week 9

Kelly Ding (she/her)

Preceptor

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Birthdays

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127.0.0.1:5000

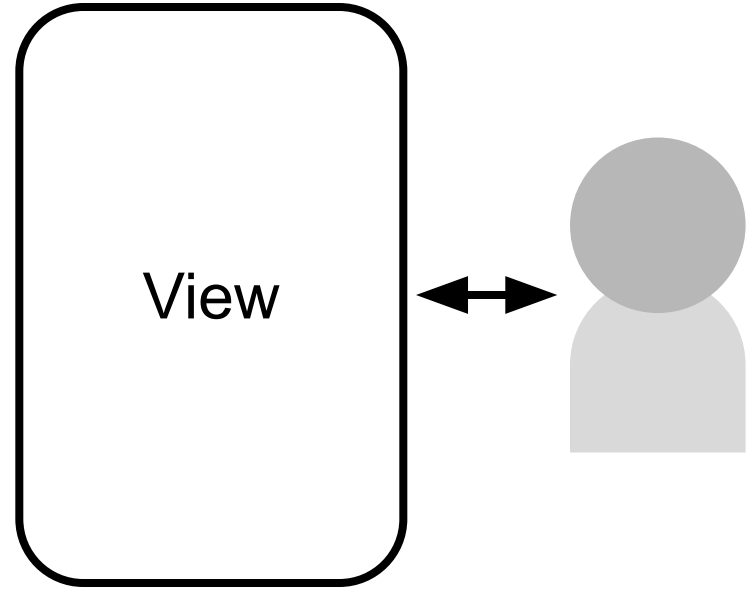
Guest

Birthdays

Add a Birthday

Add Birthday

All Birthdays



Birthdays

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127.0.0.1:5000

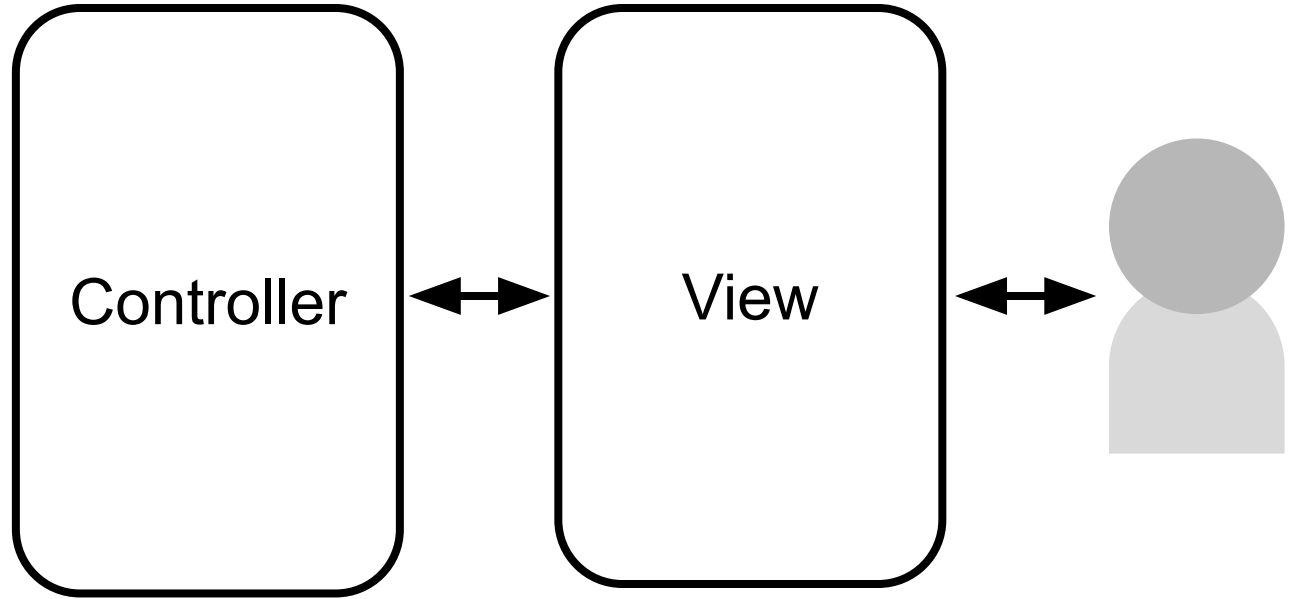
Guest

Birthdays

Add a Birthday

Add Birthday

All Birthdays



Birthdays

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127.0.0.1:5000

Guest

⋮

Birthdays

Add a Birthday

Name

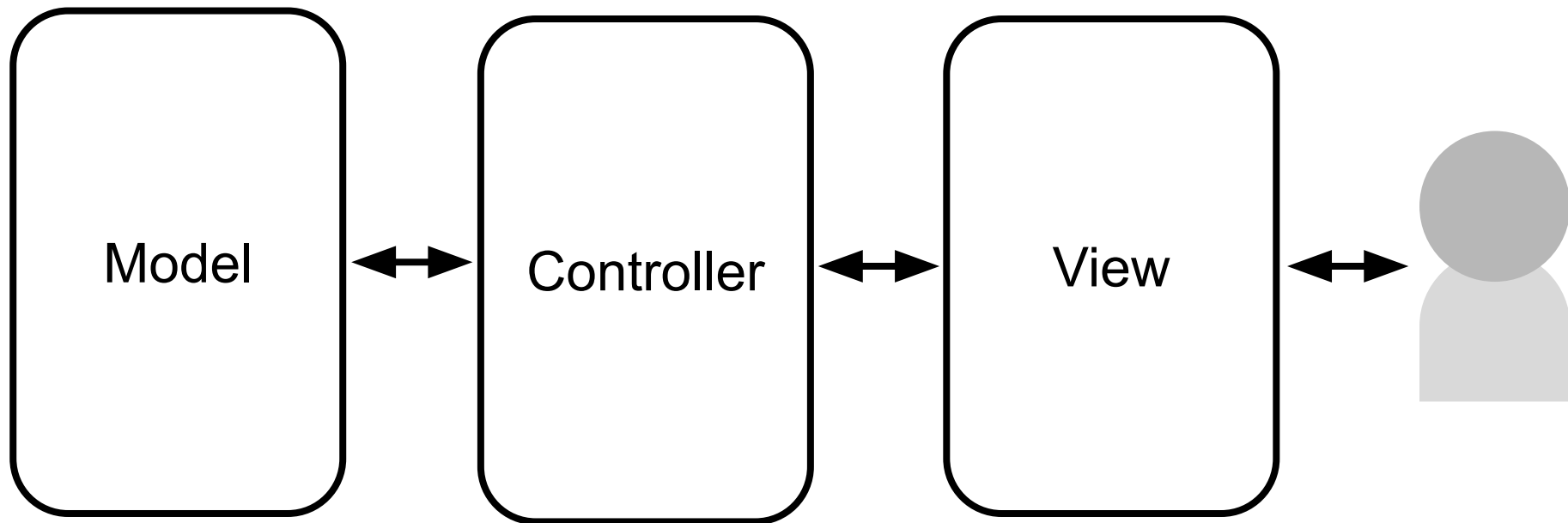
Month

Day

Add Birthday

All Birthdays

Name	Birthday
Harry	7/31
Ron	3/1
Hermione	9/19



Birthdays

127.0.0.1:5000

Guest

Birthdays

Add a Birthday

Add Birthday

All Birthdays

Name	Birthday
Harry	7/31
Ron	3/1
Hermione	9/19

birthdays/
 static/
 styles.css
 templates/
 index.html
app.py
birthdays.db

Routes

<https://example.com/>

<https://example.com/add>

```
@app.route("/")  
def index():  
    # Return index page
```

```
@app.route("/")  
def index():  
    return render_template("index.html")
```

```
flask run
```


Forms

POST vs GET Request Methods

POST

- POST is used to send data to a server to create/update a resource.

GET

- GET is used to request data from a specified resource.

A user goes to a website and
lands on the /login page.

A user goes to a website and
lands on the /login page.

GET

A user enters their username
and password and clicks Log In.

A user enters their username
and password and clicks Log In.

POST

A user submits a birthday.

A user submits a birthday.

POST

Form element

Route to request

Request method



```
<form action="/" method="post">
```

...

```
</form>
```

Elements of forms

```
<input name="friend" type="text">
```

```
<input name="month" type="number">
```

```
<button type="submit">...</button>
```

Elements of forms

```
<input name="friend" type="text">
```

```
<input name="month" type="number">
```

```
<button type="submit">...</button>
```



Submits form when clicked!

Add a form

In **index.html**, add a form to submit new birthdays. The form should have the following traits:

- The form should submit to the default **"/"** route, using the **POST** method.
- The form should include an input for a **name**, as well as the **month** and **day** of a birthday.
- The form should include a button to **submit** the form.

Updating with
`db.execute`

app.py

```
request.form.get("friend")
```

index.html

```
<input name="friend" type="text">
```

```
INSERT INTO table (column1, column2)  
VALUES (value1, value2);
```

```
db = SQL("sqlite:///birthdays.db")  
db.execute(...)
```



```
db.execute("SELECT * FROM birthdays WHERE month = ?", month)
```

```
db.execute("SELECT * FROM birthdays WHERE month = ? AND day = ?", month, day)
```

Update a database

In **app.py**, insert a new entry to the birthdays table of **birthdays.db** when the user submits data via **POST**:

- Use **request.form.get** to get the values from each named input element.
- Use **db.execute** with **?** placeholders to insert those values into the birthdays table.
- Be sure to *validate* user input. For example, check if a value is **None** before inserting the value into the database.

Template Rendering

app.py

```
render_template("index.html",  
                message="Hello")
```

index.html

```
<p>The message is: {{ message }}</p>
```

The message is: Hello

app.py

```
render_template("index.html",  
                list=["Apple", "Banana"])
```

index.html

```
{% for item in list %}  
    <p>{{ item }}</p>  
{% endfor %}
```

Apple
Banana

app.py

```
render_template("index.html",  
                bday={"month": "Feb", "day": 24})
```

index.html

```
<p>Birthday month is {{ bday.month }}</p>
```

Birthday month is Feb

{% ... %}

Jinja expressions (e.g., **for**)

{{ ... }}

Jinja variables (e.g., **bday**, **message**)

Render birthdays

In **app.py** and **index.html**, query for birthdays and display them in a table.

- Use **db.execute** to query for a list of birthdays in the birthdays table.
- Use Jinja syntax to print out birthdays one by one in a table.

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